

CAT 2025

MBA FASTRACK BATCH

Lecture - 05

Data Interpretation

Multiple Charts

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Basic

Table

Bar

Line

Pie

Casualty

Advanced.



Bubble

Scatter

Candle Stick.



TOPICS

to be covered

1

DI Sets on Multiple Charts

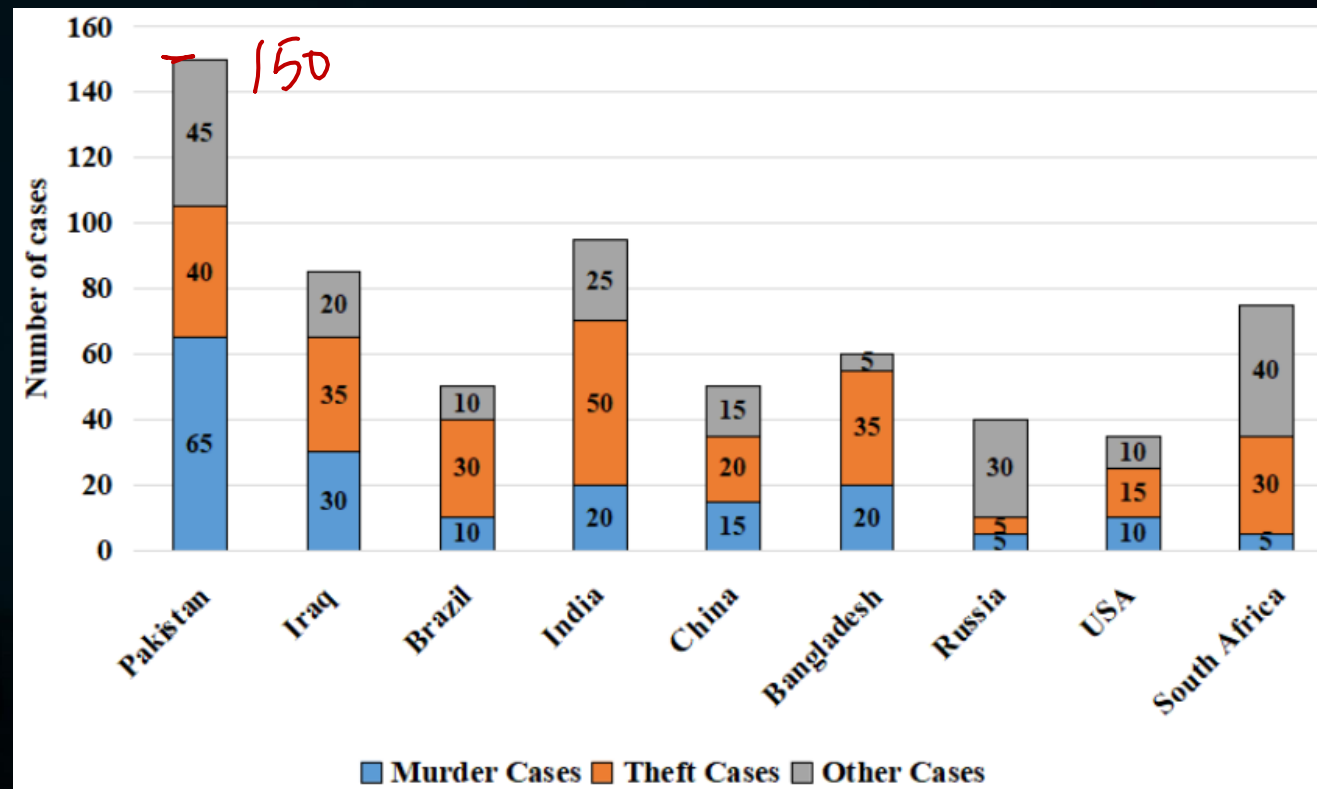


Set-1

Read the following data and answer the given questions.



The graph below depicts the crime cases registered against foreigners in different countries in a given year. The crime cases are classified into three categories: Murder cases, theft cases and other cases. The data for nine countries are shown in the figure below. These nine countries belong to the top ten countries in terms of the total number of cases registered. The tenth country is Malaysia, where only 523 theft cases were registered.



Read the following data and answer the given questions.



The adjacent table shows the rank of the ten countries in terms of the number of cases registered in each of the three categories of crimes. If two countries are assigned rank 2nd, then for any given crime, 3rd rank cannot be assigned to any country.

Countries	Murder Cases	Theft Cases	Other Cases
Pakistan	*	*	*
Iraq	*	4	*
Brazil	6	6	*
India	3	2	*
China	*	8	*
Bangladesh	3	4	9
Russia	8	10	*
USA	6	9	*
South Africa	8	6	2
Malaysia	10	*	10

A → 65 → (2)
B → 73 → (1)
C → 65 → (2)
D → 60 → (4)
E → 52 → (5)
F → 52 → (5)
G → 51 → (1)

Read the following data and answer the given questions.



Malaysia Theft = 523
Mur = 0 ✓ other = 0

65 / 563



Countries	Murder Cases	Theft Cases	Other Cases
Pakistan	1 *	3 *	1 *
Iraq	2 *	4	5 *
Brazil	6	6	7 *
India	3	2	4 *
China	5 *	8	6 *
Bangladesh	3	4	9
Russia	8	10	3 *
USA	6	9	7 *
South Africa	8	6	2
Malaysia	10	1 *	10

QUESTIONS



#Q1. The rank of China in the murder cases category is?

- A. 2nd B. 5th C. 7th D. 8th

$$\text{Murder} = 0 + 65 = \textcircled{65}$$

$$\text{Theft} = 523 + 40 = \textcircled{563}$$

#Q2. In the top two countries on the basis of total registered cases the approximate ratio of total number of murder cases to total number of theft cases is:

- A. 1:6 B. 1:7 C. 1:9 D. 1:4

#Q3. Which of the following is correct about the ranks of countries in the other crimes category?
i) USA: 7 ii) Russia: 3

- A. Only (i) is correct ~~B. Only (ii) is correct~~
C. Both (i) and (ii) are correct D. Both (i) and (ii) are incorrect

#Q4. The sum of the ranks of Pakistan in the three categories of crimes is?

- ~~A. 4~~ B. 7 C. 5 ~~D. 10~~

QUESTION- 1



#Q. The rank of China in the murder cases category is?

A. 2nd

B. 5th

C. 7th

D. 8th

QUESTION- 2



#Q. In the top two countries on the basis of total registered cases, the approximate ratio of total number of murder cases to total number of theft cases is:

A. 1:6

B. 1:7

C. 1:9

D. 1:4

QUESTION- 3



#Q. Which of the following is correct about the ranks of countries in the other crimes category?

i) USA: 7

ii) Russia: 3

A. Only (i) is correct

B. Only (ii) is correct

C. Both (i) and (ii) are correct

D. Both (i) and (ii) are incorrect

QUESTION- 4



#Q. The sum of the ranks of Pakistan in the three categories of crimes is?

A. 4

B. 7

C. 5

D. 10



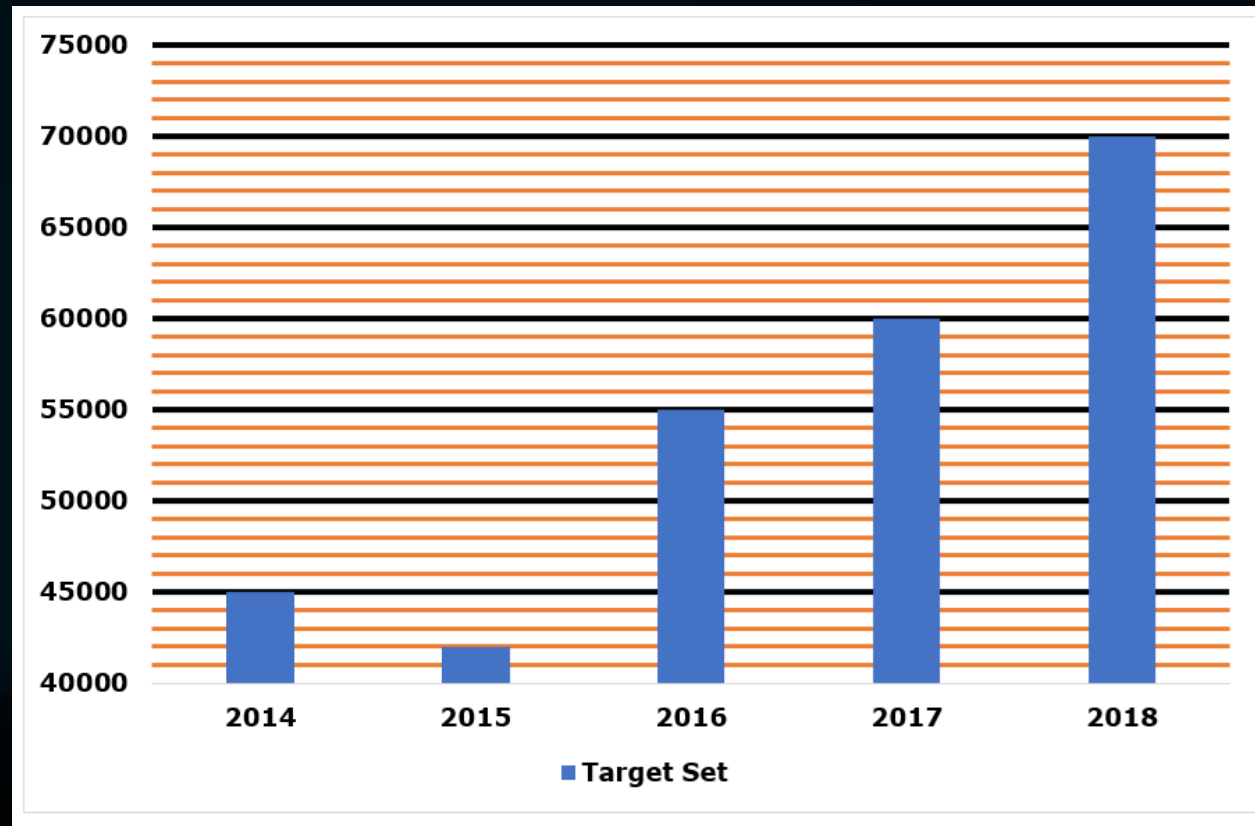
Set-2

Read the following data and answer the given questions.



An automobile company, XYZ, sets a target for the number of vehicles it is going to produce every year. The unsold vehicles are stored in inventory.

The given bar chart shows the target of vehicle production set by the company for 5 different years. (2014-2018)

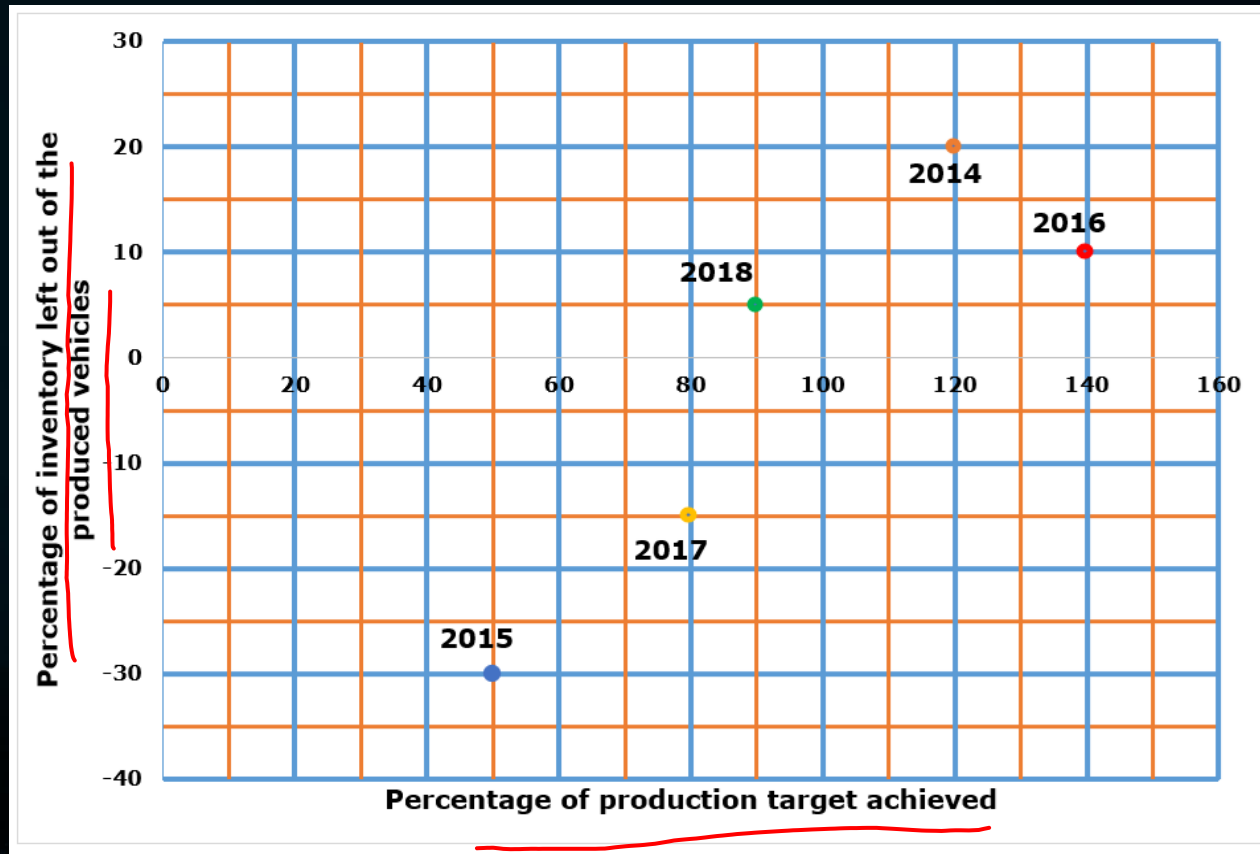


Read the following data and answer the given questions.



The given scatter graph shows the percentage of production target achieved (exceeded in some cases), along x-axis and the percentage of unsold inventory out of the produced vehicles for a particular year along y-axis.

Note: Negative inventory means that the inventory from last year was used to meet the supply for that year.



Read the following data and answer the given questions.

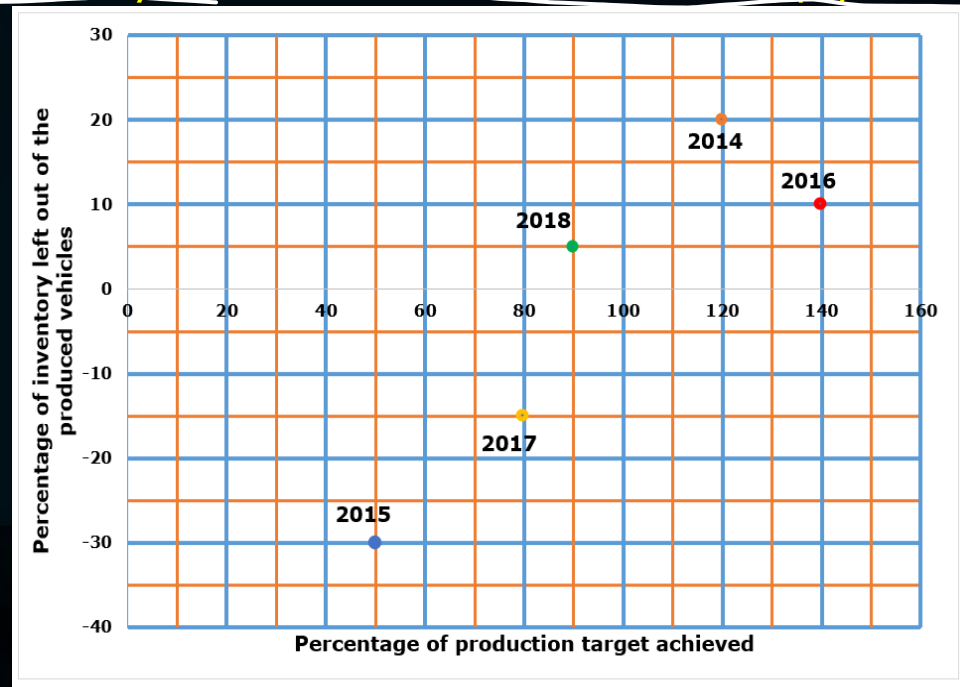
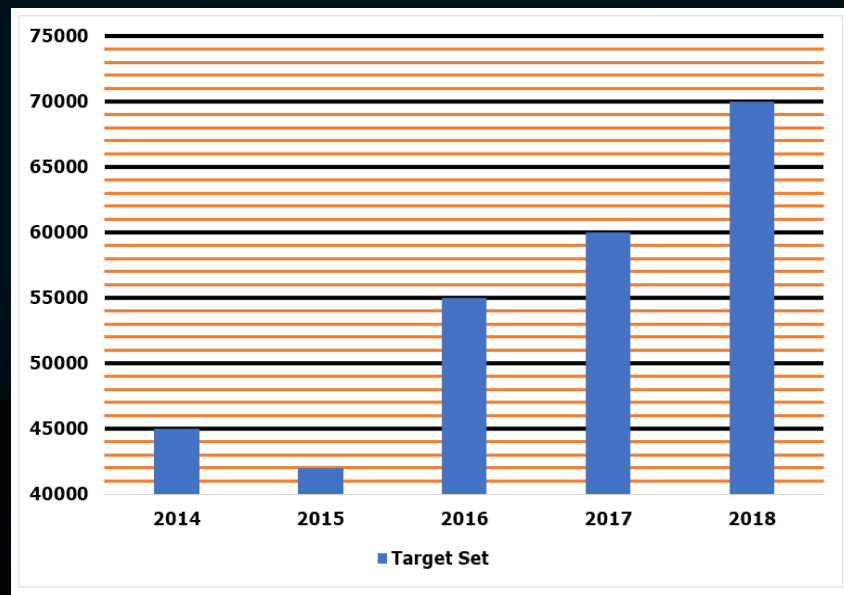


An automobile company, XYZ, sets a target for the number of vehicles it is going to produce every year. The unsold vehicles are stored in inventory.

The given bar chart shows the target of vehicle production set by the company for 5 different years. (2014-2018)

The given scatter graph shows the percentage of production target achieved (exceeded in some cases), along x-axis and the percentage of unsold inventory out of the produced vehicles for a particular year along y-axis.

Note: Negative inventory means that the inventory from last year was used to meet the supply for that year.



QUESTIONS



#Q1. What is the ratio of sum of actual production in years 2016 and 2018 to the sum of production target in years 2015 and 2017?

- A. 125:102 B. 140:69 C. 70:51 D. 125:69

#Q2. What is the total cumulative inventory left at the end of 2017?

#Q3. The actual production for the year 2014 is what percentage of the total production for all the years?

- A. 7.98% B. 29.27% C. 18.25% D. 20.53%

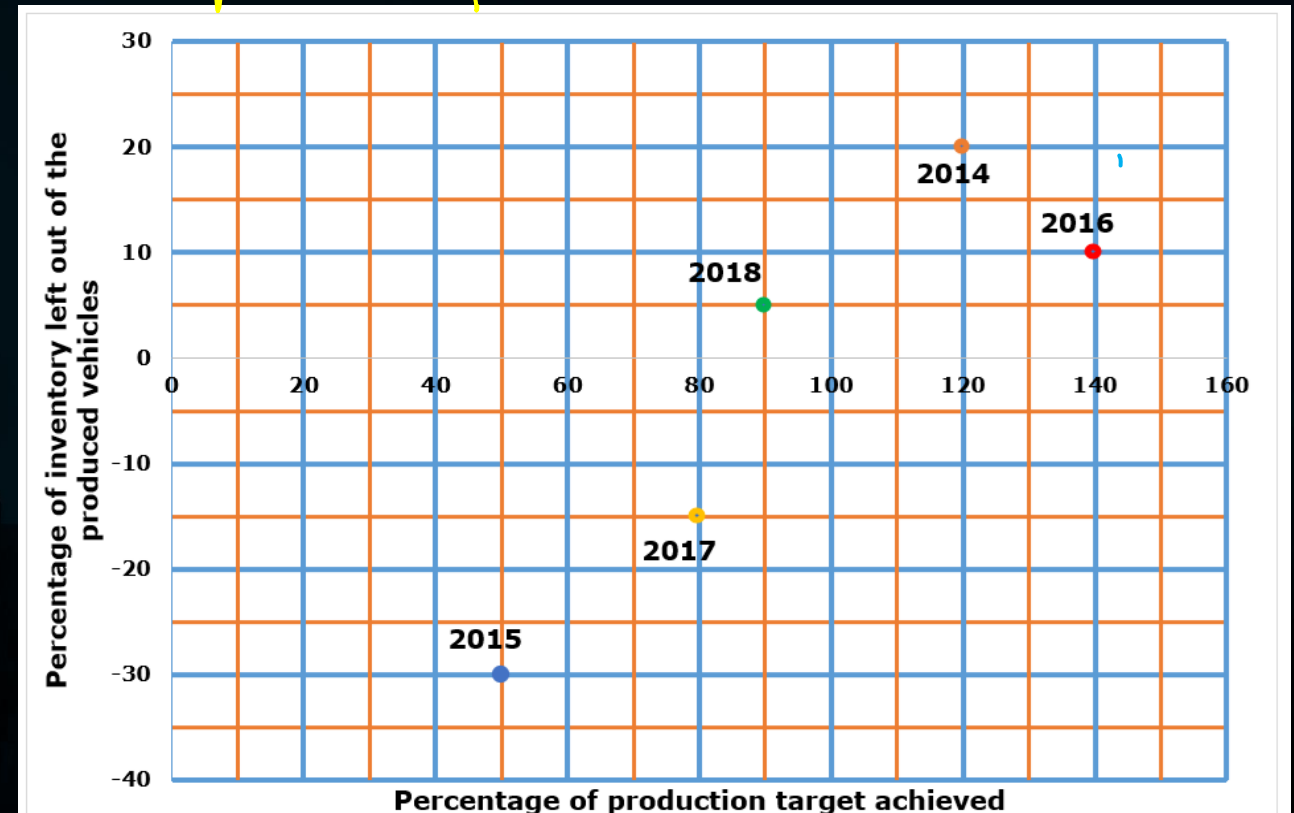
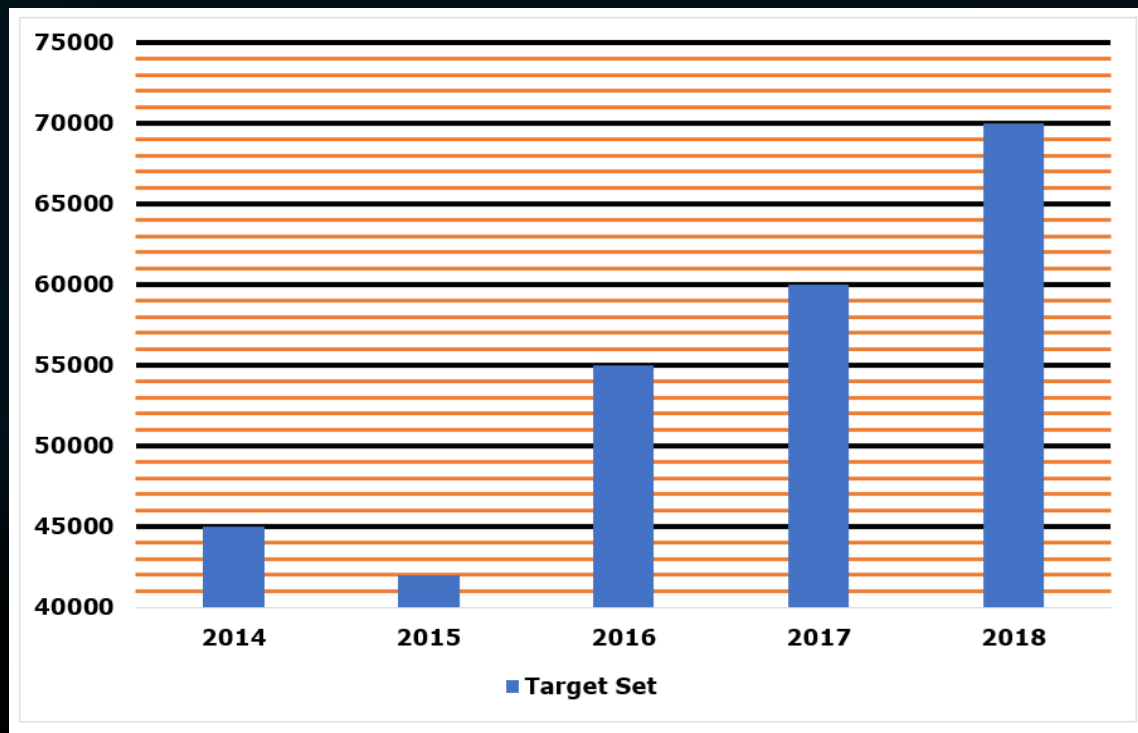
#Q4. No cumulative inventory is left at the end of 2019. If the cumulative inventory of all previous years is used in 2019, which is 12.5% of actual production in 2019, then find the actual production in year 2019.



	2014	2015	2016	2017	2018
Target	45000	42000	55000	60,000	70000
Achieved	54000	21000	77000	48000	63000
Inventory	10800	-6300	7700	-7200	3150
Cum Inc	10800	4500	12200	5000	8150

21000

23000



QUESTION-1



#Q. What is the ratio of sum of actual production in years 2016 and 2018 to the sum of production target in years 2015 and 2017?

$$77000 + 63000 = 140,000$$

$$102000$$

A. 125:102

B. 140:69

✓ C. 70:51

D. 125:69

$$\frac{140}{102} = \frac{70}{51}$$

	2014	2015	2016	2017	2018
Target	45000	42000	55000	60,000	70000
Achieved	54000	21000	77000	48000	63000
Inventory	10800	-6300	7700	-7200	3150
Cum Inv	10800	4500	12200	5000	8150

QUESTION- 2



#Q. What is the total cumulative inventory left at the end of 2017?

5000

	2014	2015	2016	2017	2018
Target	45000	42000	55000	60,000	70000
Achieved	54000	21000	77000	48000	63000
Inventory	10800	-6300	7700	-7200	3150
Cum Inv	10800	4500	12200	5000	8150

QUESTION- 3



#Q. The actual production for the year 2014 is what percentage of the total production for all the years?

- A. 7.98%
- B. 29.27%
- C. 18.25%
- ~~D. 20.53%~~

$$\frac{54000}{263000}$$

$$\frac{54}{263}$$

$$10\% \rightarrow 26.3$$

$$20\% \rightarrow 52.6$$

	2014	2015	2016	2017	2018
Target	45000	42000	55000	60,000	70000
Achieved	54000	21000	77000	48000	63000
Inventory	10800	-6300	7700	-7200	3150
Cum Inv	10800	4500	12200	5000	8150

QUESTION- 4



#Q. No cumulative inventory is left at the end of 2019. If the cumulative inventory of all previous years is used in 2019, which is 12.5% of actual production in 2019, then find the actual production in year 2019.

$$8150 = 12.5\% (\text{Prod in 2019})$$

$$\text{Prod in 2019} = 8 \times 8150 = 65200$$

	2014	2015	2016	2017	2018
Target	45000	42000	55000	60,000	70000
Achieved	54000	21000	77000	48000	63000
Inventory	10800	-6300	7700	-7200	3150
Cum Inv	10800	4500	12200	5000	8150

0/1/2/3/4



Set-3

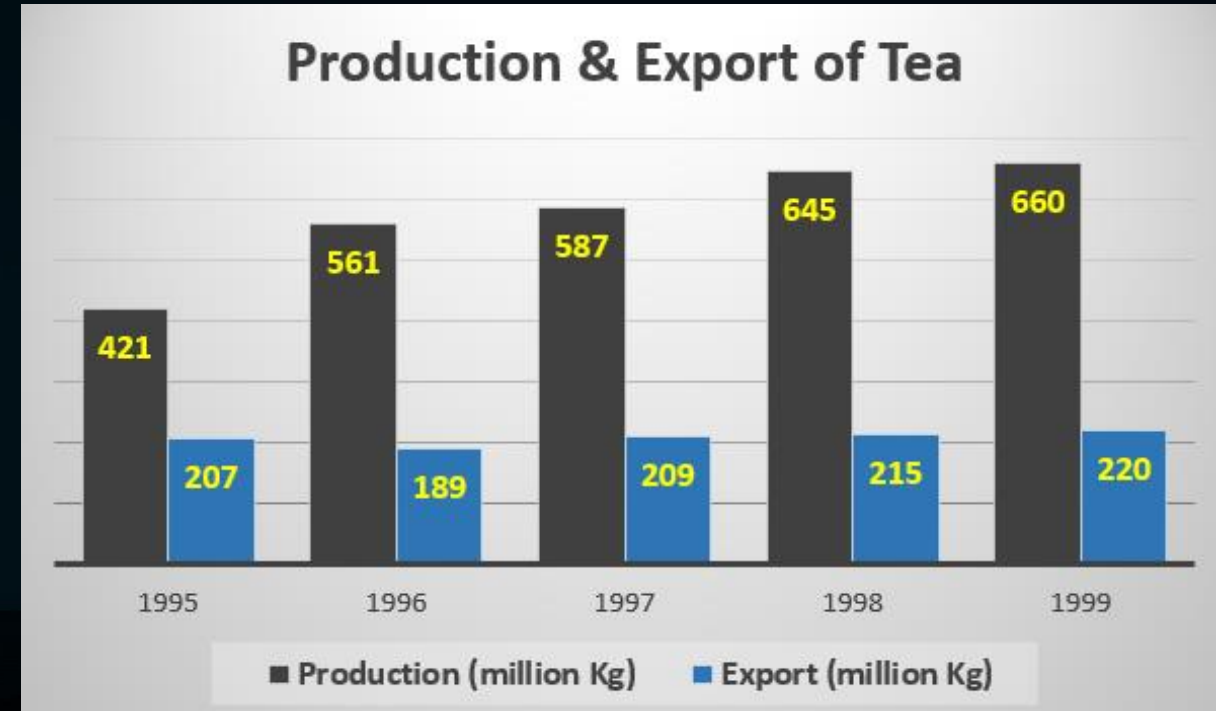
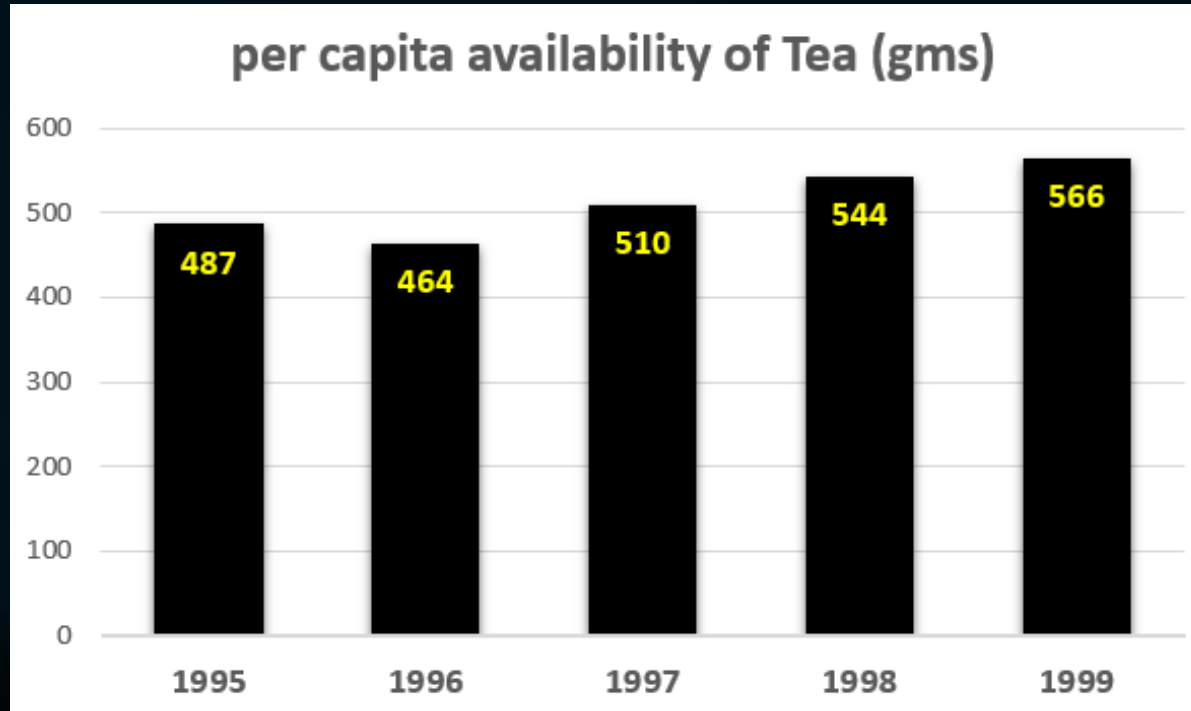
Read the following data and answer the given questions.



The data below is about the production and export of tea for a country 'Chaidesh'.

Quantity available for the people of Chaidesh = Production – Export

Per Capita Availability = (Quantity available for the people of Chaidesh) / Total Population



QUESTIONS



#Q1. In which year, between 1996–1999, was Chaidesh's export of tea, as a proportion of tea produced, the highest?

- A. 1996 B. 1997 C. 1998 D. 1999

#Q2. In which of the following years was the population of Chaidesh the lowest?

- A. 1995 B. 1996 C. 1997 D. 1999

#Q3. Which of the following options is nearest to the actual population (in millions) of Chaidesh in the year 1996?

- A. 600 B. 700 C. 800 D. 900

QUESTION-1



Both included.

#Q. In which year, between 1996–1999, was Chaidesh's export of tea, as a proportion of tea produced, the highest?

A. 1996

$$\frac{189}{561}$$

$$\frac{189}{561}$$

B. 1997

$$\frac{209}{587}$$

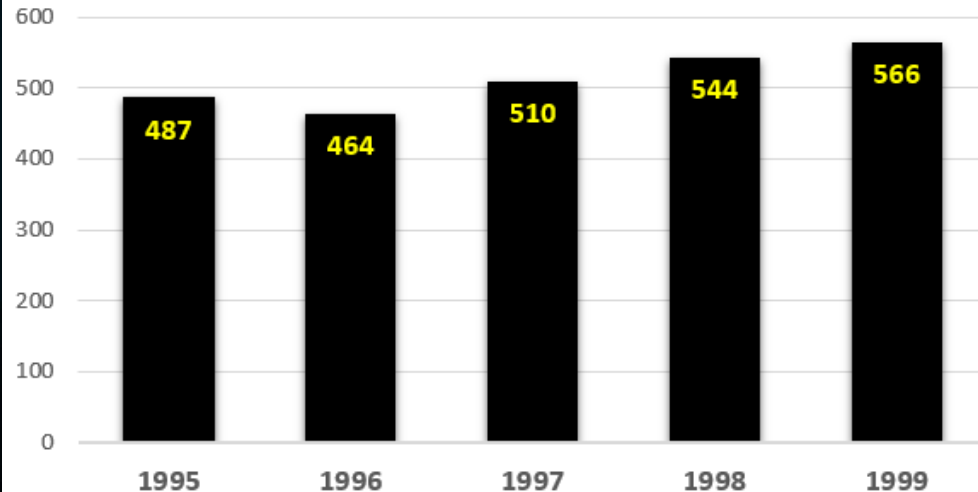
~~C. 1998~~

$$\frac{215}{645} = \frac{1}{3}$$

~~D. 1999~~

$$\frac{220}{660} = \frac{1}{3}$$

per capita availability of Tea (gms)



Production & Export of Tea



QUESTION- 2

#Q. In which of the following years was the population of Chaidesh the lowest?

A. 1995

$$\frac{214}{487}$$

↓ 1/2

B. 1996

$$\frac{372}{464}$$

↑ 1/2

C. 1997

$$\frac{378}{510}$$

↑ 1/2

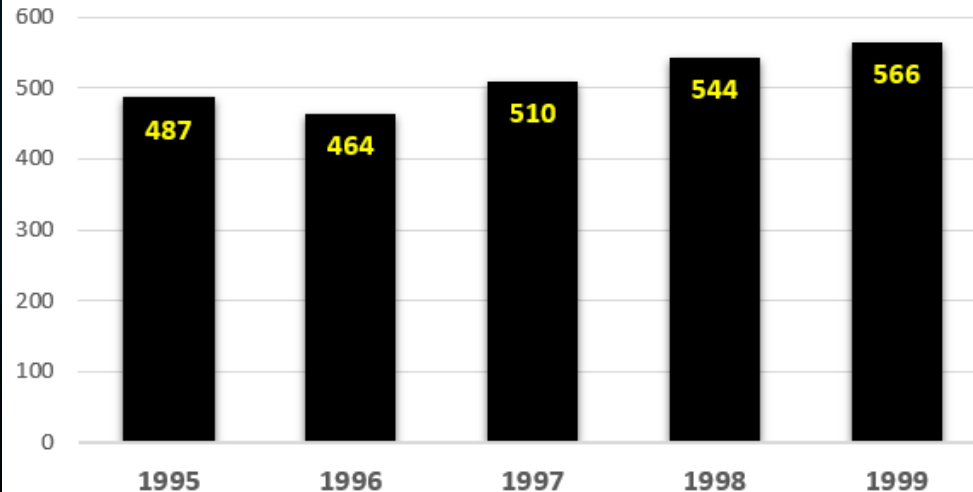
D. 1999

$$\frac{440}{566}$$

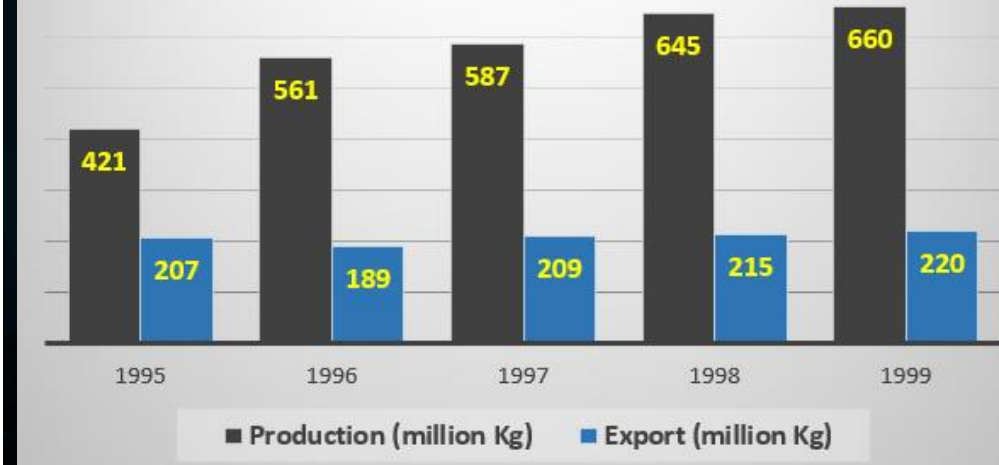
↑ 1/2

$$10^6 \times 10^3 = 10^9$$

per capita availability of Tea (gms)



Production & Export of Tea



$$PCA = \frac{Avai}{Popu}$$

$$Popu = \frac{Avai}{PCA}$$

$$Avai = Prod - Exp$$



QUESTION- 3



#Q. Which of the following options is nearest to the actual population (in millions) of Chaidesh in the year 1996?

A. 600

B. 700

✓ C. 800

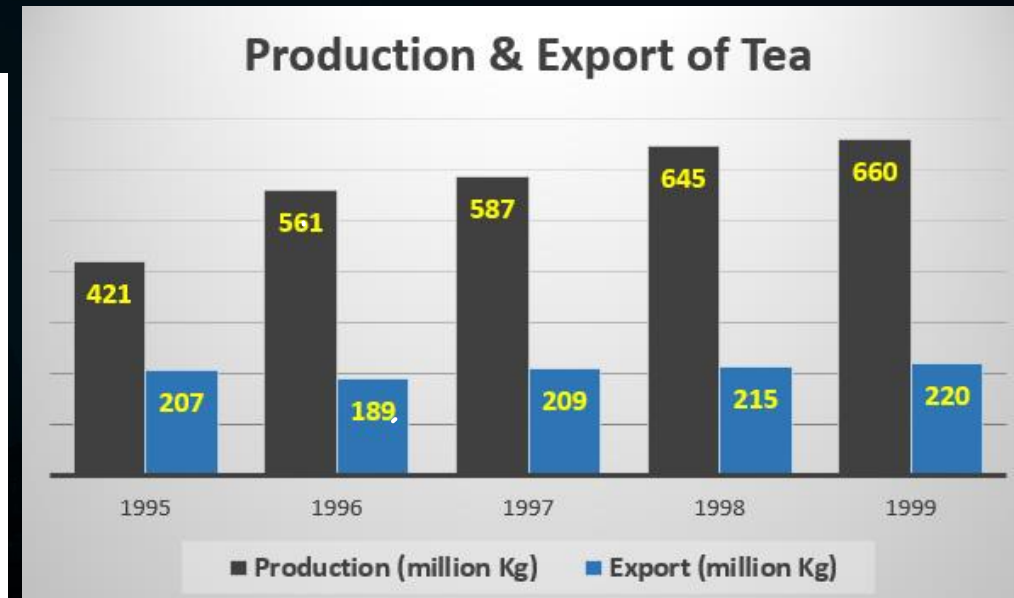
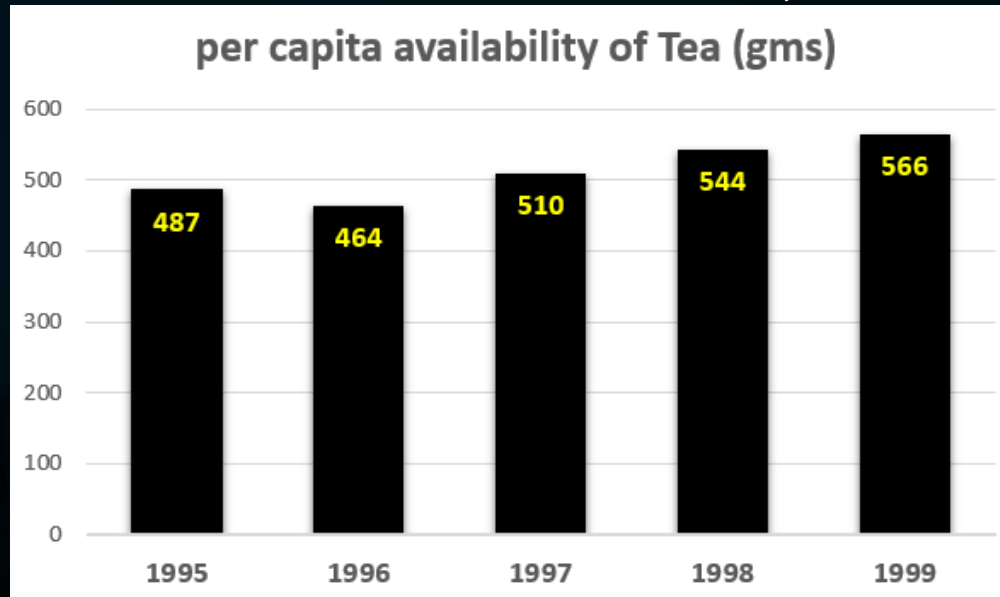
D. 900

$$\frac{372000}{464}$$

$$464000$$



$$\frac{372 \times 10^3}{464}$$





THANK
You

